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|----------------------|---------------------------------------------------------------|
| <b>Date</b>          | 03 July 2026                                                  |
| <b>Team ID</b>       | SWTID-2026-4593                                               |
| <b>Team Size</b>     | 3                                                             |
| <b>Team Leader</b>   | Hari Bellamkonda                                              |
| <b>Team Member 1</b> | Kesha Vardhan Chinni                                          |
| <b>Team Member 2</b> | Akili Venkata Siva Sai Srikar                                 |
| <b>Project Name</b>  | Machine Learning–Based Credit Card Approval Prediction System |
| <b>Maximum Marks</b> | 4 Marks                                                       |

## Define Problem Statements

### 1. Problem Statement

Financial institutions receive thousands of credit card applications every day. The traditional approval process mainly depends on manual verification of applicant information such as income, employment status, credit history, financial background, and other eligibility factors. Manual evaluation requires considerable time and human effort, often resulting in delayed decisions, inconsistent assessments, and increased operational costs. There is a need for an intelligent and automated solution that can accurately predict credit card approval using Machine Learning techniques while supporting faster and more reliable decision-making.

### 2. Existing System

| Aspect                 | Description                               |
|------------------------|-------------------------------------------|
| Application Evaluation | Manual verification by banking staff      |
| Decision Making        | Human judgement                           |
| Processing Time        | Several hours or days                     |
| Consistency            | Varies between evaluators                 |
| Scalability            | Limited for large numbers of applications |
| Operational Cost       | High                                      |
| Risk of Human Error    | High                                      |
| Automation             | Minimal                                   |

### 3. Problems Identified in the Existing System

- Time-consuming verification process.
- Heavy dependency on manual decision-making.
- Inconsistent approval decisions.
- Increased operational costs.
- Difficulty processing large application volumes.
- Higher probability of human error.
- Limited scalability.
- Slow customer response.

### 4. Root Cause Analysis

| Problem                  | Root Cause                          |
|--------------------------|-------------------------------------|
| Slow processing          | Manual verification of applications |
| Inconsistent decisions   | Human judgement varies              |
| High operational cost    | Large workforce involvement         |
| Customer dissatisfaction | Long waiting time                   |
| Poor scalability         | Increasing number of applications   |

#### Note

The identified problems highlight the need for an intelligent, automated, and data-driven credit card approval prediction system capable of improving efficiency, consistency, and decision accuracy.

## 5. Proposed Solution

The proposed Machine Learning–Based Credit Card Approval Prediction System automates the evaluation of credit card applications using historical applicant data and trained Machine Learning models. The system analyses applicant information, predicts whether the application is likely to be approved or rejected, and provides rapid, consistent, and reliable decision support for financial institutions.

## 6. Project Objectives

1. Automate the credit card approval prediction process.
2. Reduce manual verification effort.
3. Improve prediction accuracy using Machine Learning.
4. Minimize processing time.
5. Assist financial institutions in decision-making.
6. Improve customer experience.
7. Generate reliable prediction reports.
8. Build a scalable web-based prediction platform.

## 7. Project Scope

The scope of this project includes applicant registration, secure authentication, credit card application submission, data preprocessing, Machine Learning prediction, approval result generation, database management, prediction history, administrative reporting, and analytics. The system focuses on predicting credit card approval based on applicant information while providing an intuitive web interface for both applicants and administrators.

## 8. Expected Benefits

| Benefit                      | Description                                                  |
|------------------------------|--------------------------------------------------------------|
| Fast Prediction              | Provides approval results within seconds.                    |
| Higher Accuracy              | Uses Machine Learning algorithms for intelligent prediction. |
| Reduced Cost                 | Minimizes manual processing effort.                          |
| Consistency                  | Produces standardized prediction results.                    |
| Improved Customer Experience | Faster application processing.                               |
| Scalability                  | Supports increasing numbers of applications.                 |
| Better Decision Support      | Assists administrators with data-driven decisions.           |

## 9. Conclusion

### Conclusion

The Machine Learning–Based Credit Card Approval Prediction System addresses the limitations of traditional credit card approval processes by introducing intelligent automation, faster prediction, improved accuracy, enhanced scalability, and reliable decision support. The proposed solution contributes to more efficient banking operations while improving the overall customer experience.